similarly the size of the priority queue can be bound[23]. Therefore, with a wise choice of εMAX, the **average** running time can be as low as `O(n*log(n))`, for a dataset with `n` points.

The original paper contains a more formal description of the algorithm and the theory behind it, and an automated procedure to build a hierarchical clustering structure from the reachability plot, and it's a good starting point if you'd like to deepen your understanding of this algorithm.

Another interesting reading is the paper on *De.Li.Clu.*[24], an advanced algorithm that extends OPTICS with ideas from single-linkage clustering, allowing to avoid the parameter εMAX altogether, and at the same time optimizing the algorithm and improving its running time.

12.5 Evaluating Clustering Results: Evaluation Metrics

At this point in the chapter, we have learned about three different (and increasingly complex and effective) clustering algorithm, their strengths and their weaknesses; despite being so different from one another, there is one thing that all of these algorithms have in common: they require a human to set one or more hyper-parameters in order to achieve the best result.

Setting these parameters manually can be beyond challenging: while with 2-D datasets it's possible to take a look at the resulting clustering and decide if it looks good or